Module 64.1: Asset-Backed Security (ABS) Instrument and Market Features

SCHWESERNOTES - BOOK 3

LOS 64.a: Describe characteristics and risks of covered bonds and how they differ from other asset-backed securities.

Covered bonds are senior debt obligations of financial institutions that are similar to ABSs. However, the underlying assets (the cover pool), while segregated from other assets of the issuer, remain on the balance sheet of the issuing corporation (i.e., no SPE is created). Covered bonds are issued primarily by European, Asian, and Australian banks, and the cover pool typically consists of mortgage loans (though other types of assets can be used).

In the event of issuer default, covered bond investors have the **dual recourse** of claims on both the cover pool and, in contrast to a securitization involving an SPE, claims over other assets of the issuers that have not been pledged as collateral for other debt (referred to as unencumbered assets). To mitigate credit risk faced by covered bond investors, the mortgages that make up the cover pool are subject to upper limits on loan-to-value ratios, and the value of the collateral is usually higher than the face value of covered bonds issued (referred to as overcollateralization). The cover pool is monitored by a third party to ensure adherence to these conditions. These credit enhancements, along with dual recourse, result in covered bonds generally having lower yields than comparable ABSs.

Because the cover pool remains on the balance sheet of the issuer, the issuer will not benefit from any reduction in required capital reserves that would occur under a securitization.

Unlike an ABS, in which the pool of assets is fixed at issuance, a covered bond requires the issuer to replace or augment nonperforming or prepaid assets in the cover pool so that it always provides for the covered bond's promised interest and principal payments. Covered bonds typically are not structured with credit tranching.

Covered bonds may have different provisions in case their issuer defaults. A **hard-bullet covered bond** is in default if the issuer fails to make a scheduled payment, leading to the acceleration of payments to covered bondholders. A **soft-bullet covered bond** may postpone the originally scheduled maturity date by as much as a year, should a payment on the covered bond be missed—effectively postponing default and associated payment acceleration. A **conditional pass-through covered bond** converts to a pass-through bond on the maturity date if any payments remain due, meaning that any payments subsequently recovered on the cover pool are passed through to investors.

LOS 64.b: Describe typical credit enhancement structures used in securitizations.

Internal credit enhancements of ABSs are features of the structure designed to mitigate the credit risk faced by investors due to defaults in the collateral pool. They take three main forms: overcollateralization, excess spread, and subordination (or credit tranching).

Overcollateralization occurs when the value of the collateral exceeds the face value of the ABS. For example, if the value of the collateral is $600 million and the face value of the ABS issued is $500 million, then there is $100 million overcollateralization. The collateral could experience default of up to 100 / 600 = 16.7% of value before investors in the ABS begin to suffer credit losses.

An **excess spread** feature builds up reserves in the ABS structure by earning higher income on the collateral than the coupon promised to ABS investors. This income can then be used to absorb credit losses in the collateral.

With **credit tranching** (also called **subordination**), the ABS is structured with multiple classes of securities (referred to as **tranches**), each with a different priority of claims to the cash flows of the collateral. Tranches ranked as subordinated absorb credit losses first (up to their principal values), thereby providing protection to senior tranches against credit losses. The level of protection for the senior tranche increases with the proportion of subordinated bonds in the structure.

Let's look at an example to illustrate how a senior/subordinated structure redistributes the credit risk compared to a single-class structure. Consider an ABS with the following bond classes:

Tranche Name	Face Value ($)	Interest Rate
Tranche A senior notes	300,000,000	MRR + 0.5%
Tranche B subordinated notes	80,000,000	MRR + 1.5%
Tranche C subordinated notes	30,000,000	Variable
Total	410,000,000	

Tranche C is first to absorb any losses, because it is the most junior tranche, until losses exceed $30 million in principal. Being the lowest-ranking tranche, it has a residual claim to any value left over after other, more senior tranches have been paid—and for this reason, it is often referred to as the **equity tranche**. Any losses from default of the underlying assets greater than $30 million, and up to $110 million, will be absorbed by subordinated Tranche B. The senior Tranche A is protected from any credit losses of $110 million or less, and therefore it will have the highest credit rating and offer the lowest yield of the three bond classes. It is through this credit tranching, and the bankruptcy remote nature of the SPE, that senior tranches of ABSs can receive credit ratings higher than that of the originating company that sells the collateral to the SPE.

Professor's Note

This subordination structure is not new to us—it's the same idea as the senior/junior debt and equity capital structure of a corporate issuer.

This structure is also called a **waterfall** structure because in liquidation, each subordinated tranche would receive only the "overflow" from the more senior tranche(s) if they are repaid their principal value in full.

As long as a tranche remains outstanding, it will receive its coupon payment. For example, say an investor purchases $10 million face value of the Tranche B notes. If the MRR is 4%, and if there are no defaults in the underlying collateral, the investor will receive an annual coupon of $10,000,000 × (0.04 + 0.015) = $550,000. If defaults in the collateral pool amount to $50 million, the first $30 million of losses will be allocated to Tranche C, while the remaining losses of $20 million will reduce the outstanding face value of the Tranche B note from $80 million to $60 million. The coupon earned by the investor in Tranche B notes, in this case, is equal to $550,000 × (60 / 80) = $412,500.

LOS 64.c: Describe types and characteristics of non-mortgage asset-backed securities, including the cash flows and risks of each type.

In addition to those backed by mortgages, there are ABSs backed by various types of financial assets including business loans, accounts receivable, or automobile loans. In fact, any asset that generates a future stream of cash flows could be used as collateral for securitization, including music royalties or franchise license payments. Each of these types of ABS has different risk characteristics, and their structures vary to some extent as well. Here, we explain the characteristics of two types: ABSs backed by credit card receivables and ABSs backed by loans for homeowners to install solar panels on their property (referred to as a residential solar ABS).

A key distinction between the debt-based assets acting as collateral for these ABSs is whether it is amortizing or nonamortizing. Recall that an amortizing loan has a principal balance that is scheduled to be paid down over the life of the loan. A nonamortizing loan has no such schedule for repayment of principal.

Credit Card ABS

Credit card receivable-backed securities are ABSs backed by pools of credit card debt owed to banks, retailers, travel and entertainment companies, and other credit card issuers.

The cash flow to a pool of credit card receivables includes finance charges (i.e., interest), membership and late payment fees, and principal repayments. Credit card receivables are nonamortizing loans; however, borrowers can choose to repay principal at their discretion. Interest rates on credit card receivables can be fixed or floating, typically subject to a cap (i.e., a maximum rate that the credit card lender can charge).

A credit card securitization structure will typically include a **lockout period** or **revolving period** during which ABS investors only receive interest and fees paid on the collateral. If the underlying credit card holders make principal payments during the lockout period, these payments are used to purchase additional credit card receivables, keeping the overall value of the pool relatively constant.

Professor's Note

> In our reading on mortgage-backed securities, we will discuss prepayment risk at length. Prepayment risk occurs when borrowers repay loans at a different speed to that originally anticipated by investors. During the revolving period of a credit card ABS, investors are not subject to prepayment risk because any principal payments in the underlying collateral are used to make additional loans.

Once the lockout period ends, the **amortization period** of the ABS begins, and principal payments made on the underlying collateral are passed through to security holders. Credit card ABSs typically have an early (rapid) amortization provision that provides for earlier amortization of principal when it is necessary to preserve the credit quality of the securities, akin to an acceleration of payment for a debt issuer when credit conditions worsen.

Solar ABSs

Solar ABSs are backed by loans to homeowners wishing to finance the installation of solar energy systems to reduce their energy bills. These loans are offered by specialist finance companies or the financing subsidiaries of solar energy companies.

Solar ABSs are attractive to investors focused on environmental, social, and governance (ESG) factors because, through investing in the solar ABS, they are providing funds for homeowners to switch to a renewable energy source. Solar loans themselves can be secured on the solar energy system itself or on the homeowner's property as a junior mortgage, providing extra security to ABS investors.

Internal credit enhancement methods such as overcollateralization, excess spread, or subordination are common in these structures. Solar loans are usually made to homeowners with good credit scores, who are saving on energy bills through installing the solar energy system; hence, credit risk is likely to be low. However, solar ABSs are a relatively new asset class and are yet to be tested through a full credit cycle.

Many solar ABSs have a **pre-funding period**, which allows the trust to make investments in solar-related loans for a fixed time period after raising funds through issuing the ABS. A pre-funding period allows the ABS structure to invest in a larger, more diversified pool of solar-related investments.

LOS 64.d: Describe collateralized debt obligations, including their cash flows and risks.

A **collateralized debt obligation (CDO)** is a structured security issued by an SPE for which the collateral is a pool of debt obligations. When the collateral securities are corporate and emerging market debt, they are called collateralized bond obligations (CBOs). Collateralized loan obligations (CLOs) are supported by a portfolio of leveraged bank loans. Unlike the ABSs we have discussed, CDOs do not rely solely on payments from a static collateral pool to meet obligations from issuing securities. What sets CDOs apart from regular ABSs is that CDOs have a **collateral manager** who dynamically buys and sells securities in the collateral pool to generate the cash to make the promised payments to investors.

> **Professor's Note**
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> Leveraged loans are senior secured bank loans made to companies that have high levels of debt already, or have a poor credit history.

CDOs issue subordinated tranches in a similar fashion to ABSs. In creating a CDO, the structure must be able to offer an attractive return on the lowest-ranked equity tranche, after accounting for the required yields on the more senior CDO debt tranches.

Before the 2007-2009 global financial crisis, CDOs were based on a wide variety of underlying collateral. Since the crisis, CDO structures have become less complex, and more stringent requirements have been placed on the quality of the collateral. Collateral today comprises mostly leveraged loans; hence, the most common form of CDO is the CLO.

Three major types of CLOs are as follows:

1. *Cash flow CLOs*: Payments to CLO investors are generated through cash flows on the underlying collateral.
2. *Market value CLOs*: Payments to CLO investors are generated through trading the market value of the underlying collateral.
3. *Synthetic CLOs*: The collateral pool exposure is generated through credit derivative contracts. In this type of CLO, the CLO trust does not take ownership of the collateral.

The collateral of a CDO is subject to a series of prespecified tests to protect CLO investors from default:

- Coverage of payment obligations to the CLO investors by cash flows from the collateral
- Overcollateralization levels for each tranche—breaches of overcollateralization limits cause cash flows to be redirected to purchase additional collateral or to pay off the senior-most tranches of the CDO

- Diversification in the collateral pool
- Limitations on the amount of CCC rated debt in the collateral pool

Reading 64: Key Concepts

SCHWESERNOTES - BOOK 3

LOS 64.a

Covered bonds are similar to asset-backed securities, but instead of creating an SPE, the collateral (cover pool) remains on the balance sheet of the issuer. Covered bonds give bondholders recourse to the issuer as well as the asset pool, which increases the bonds' credit quality.

LOS 64.b

Internal credit enhancements of ABS include the following:

- *Overcollateralization*. The value of the collateral is greater than the face value of the ABS securities.
- *Excess spread*. Income from the collateral in excess of the payment obligations to ABS investors acts as a reserve to absorb default losses in the collateral pool.
- *Credit tranching*. Credit losses are first absorbed by the tranche with the lowest priority, and after that, by any other subordinated tranches, in order.

LOS 64.c

Credit card ABSs are backed by credit card receivables, which are nonamortizing receivables. Credit card ABSs typically have a lockout/revolving period during which only interest and fees are passed through from the collateral pool to investors, while any principal payments on the receivables are used to purchase additional receivables. Following this period, the ABSs have an amortization period where principal payments are passed through to ABS investors.

Solar ABSs are backed by loans to homeowners to finance installation of domestic solar energy equipment. Solar loans can be secured on the solar energy equipment or the homeowner's residence. Solar ABSs may be attractive to ESG investors.

LOS 64.d

Collateralized debt obligations (CDOs) are structured securities backed by a pool of debt obligations that are managed by a collateral manager. CDOs include collateralized bond obligations (CBOs) backed by corporate and emerging market debt, and collateralized loan obligations (CLOs) backed by leveraged bank loans.

Three major types of CLO are as follows:

- *Cash flow CLOs*. Payments to CLO investors are generated through cash flows on the underlying collateral.
- *Market value CLOs*. Payments to CLO investors are generated through trading the market value of the underlying collateral.
- *Synthetic CLOs*. Collateral pool exposure is generated through credit derivative contracts.